THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2241
Introduction to Mathematical Analysis
Tutorial 2

Problem 1. Compute the limits:

(a) $\lim_{n\to\infty}\left(\frac{1}{2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{n\cdot(n+1)}\right)$

(b) $\lim_{n\to\infty}\sqrt[n]{\left(1-\frac{1}{n}\right)}$

(c) $\lim_{n\to\infty}\left(\frac{1}{n^2}+\frac{1}{(n+1)^2}+\cdots+\frac{1}{(2n)^2}\right)$

(d) $\lim_{n\to\infty}\left(\frac{1}{\sqrt{(n^2+1)}}+\frac{1}{\sqrt{(n^2+2)}}+\cdots+\frac{1}{\sqrt{(n^2+n)}}\right)$

Problem 2. Let $a_1, a_2, \ldots, a_m$ be m positive numbers. Prove that

$$\lim_{n\to\infty}\sqrt[n]{a_1^n+a_2^n+\cdots+a_m^n}=\max\{a_1,a_2,\cdots,a_m\}.$$

Problem 3. (a) Let $\{a_n\}$ and $\{b_n\}$ be sequences. Prove that if $\lim_{n\to\infty} a_n = 0$ and if $\{b_n\}$ is bounded, then

$$\lim_{n\to\infty} a_n b_n = 0.$$

(b) Suppose now that $\lim_{n\to\infty} a_n = L \neq 0$ and that $\{b_n\}$ is bounded. What can we conclude about the convergence of the sequence $\{a_n b_n\}$?

Problem 4. Suppose $x_1 = \sqrt{2}$, $x_{n+1} = \frac{1}{x_n+2}$, $(n = 1, 2, 3\ldots.)$. Prove that the sequence $\{x_n\}$ is convergent, and find the limit.
Hint: If $0 < x_n < \sqrt{2}-1$, *then* $x_{n+1} > \sqrt{2}-1$; *if* $x_n > \sqrt{2}-1$, *then* $0 < x_{n+1} < \sqrt{2}-1$.

Problem 5. (a) Suppose $a_n > 0$, $(n = 1, 2, \cdots)$, and $\lim \frac{a_n}{a_{n+1}} = l > 1$. Prove that $\lim a_n = 0$.

(b) Prove that the limit $\lim_{n\to\infty} \frac{n^2}{3^n}$ exists and find the limit.

Food for Thought. Find the limits $\lim_{n\to\infty} \frac{3^n}{n!}$, $\lim_{n\to\infty} \frac{n!}{n^n}$ and compare the 'ultimate behaviour' of n^a, b^n, $n!$ and n^n, $(a, b > 1)$.

Problem 6. Find the limits:

(a)

$$\lim_{n\to\infty}\frac{\sum_{p=1}^{n} p!}{n!}$$

(b)

$$\lim_{n\to\infty}\left((n+1)^\alpha - n^\alpha\right), \quad 0 < \alpha < 1$$